OOP-Based Library Management System

Advanced Programming Project | B.Tech Semester 4

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Project Objectives

Build a GUI

Developgraphical interface for library inventory management

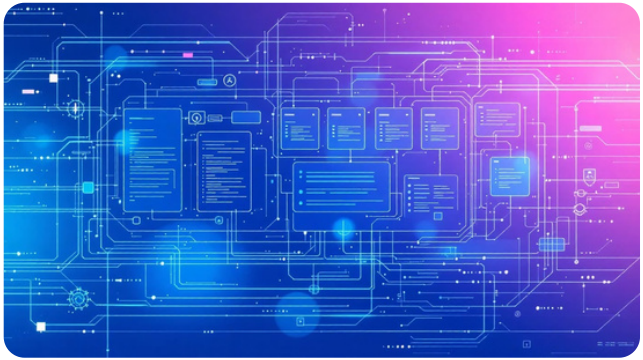
Apply OOP Principles

Implement coreJavaconceptsin real-world logic

Modular Structure

Organizecodeusingpackage-based architecture

Project Architecture



Core Components

- src/LibraryItem.java:Abstract Parent Class
- src/Book.java: Inheritance Subclass
- src/LibraryGUI.java: Swing UI Entry Point
- docs/: System documentation
- report/: Analysis folders

Key OOP Principle: Abstraction

What is Abstraction?

Hidingcompleximplementation details whilst exposing only essential functionality to users.

Implementation in Project

LibraryItem isanabstractclassdefining that every item must have an ID and Title.

Cannot be instantiated directly4ensures data integrity through structure.

Code Snippet

```
abstract class LibraryItem {
protected String id;
protected String title;

 // Abstract method
 abstract void displayInfo();
}
```

Key OOP Principle: Inheritance

1

Parent Class

LibraryItemprovidescommon attributes: ID, Title

2

Child Class

BookextendsLibraryItem, inherits properties

3

Added Features

Bookaddsspecificdata: Author, ISBN

Key OOP Principle: Encapsulation

1

Data Hiding

Private fieldsprevent direct access to internal data

2

Access Methods

Public gettersandsetters control data modification

3

Data Integrity

Validationlogicensures consistent state

Technical Implementation

Java Swing GUI

- JTable with DefaultTableModel
- Real-time data display
- Event handling for user actions
- "Add Item" button processing

Execution & Deployment

01

Compilation

`javac src/*.java` compiles all source f iles

02

Run Application

`java src.LibraryGUI` launches the interface

03

Requirements

JavaDevelopmentKit (JDK) 11 or higher installed

Project Highlights

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OOP Principles
Abstraction, Inheritance, Encapsulation

100%

No external libraries required

1

Modular Design
Package-based structure for scalability

Future Scope & Enhancements

Database Integration

ConnecttoMySQLforpersistent storage

Advanced Search

Implementqueryfunctionalityfor efficient item retrieval

Additional Features

- User authentication system
- Borrowing/return tracking
- Reporting and analytics